Experiment 10: Map Coloring using CSP

Aim

To implement the Map Coloring problem using Constraint Satisfaction Problem (CSP).

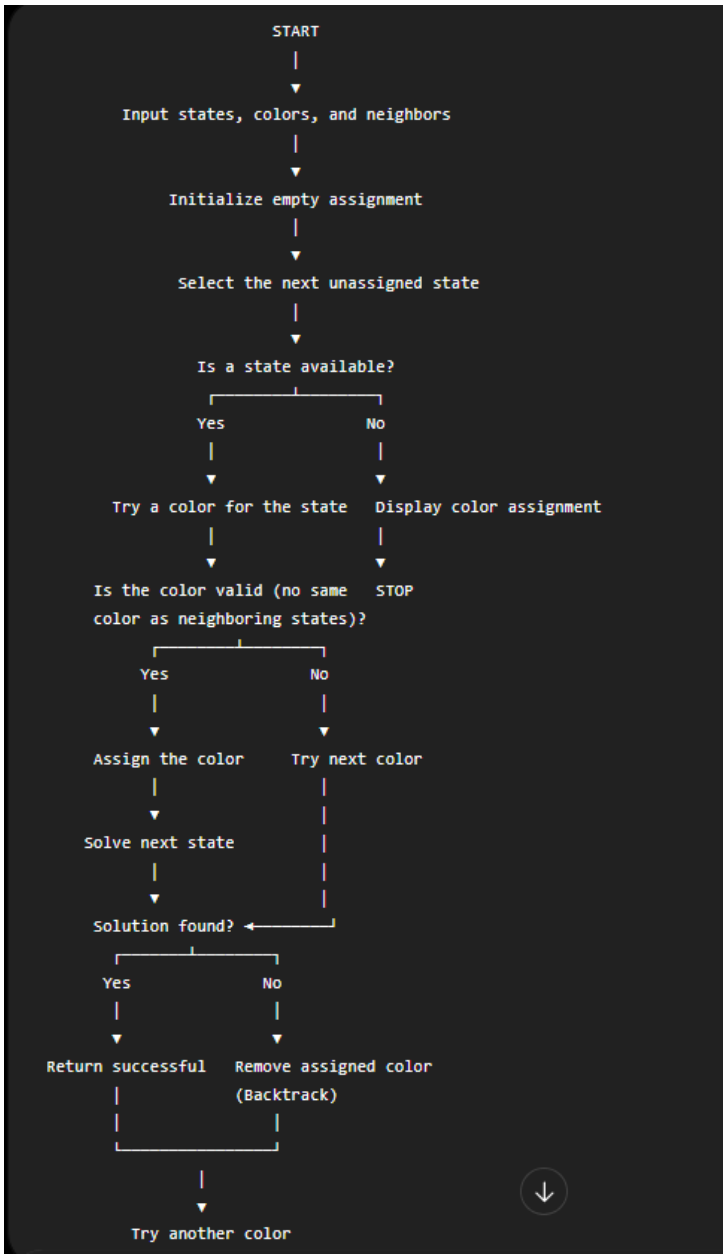
Objective

- To assign colors to regions.
- To ensure adjacent regions have different colors.

Algorithm

1. Create the map and adjacency list.
2. Define available colors.
3. Assign colors recursively.
4. Check whether neighboring regions have different colors.
5. If valid, continue.
6. Display the final coloring.

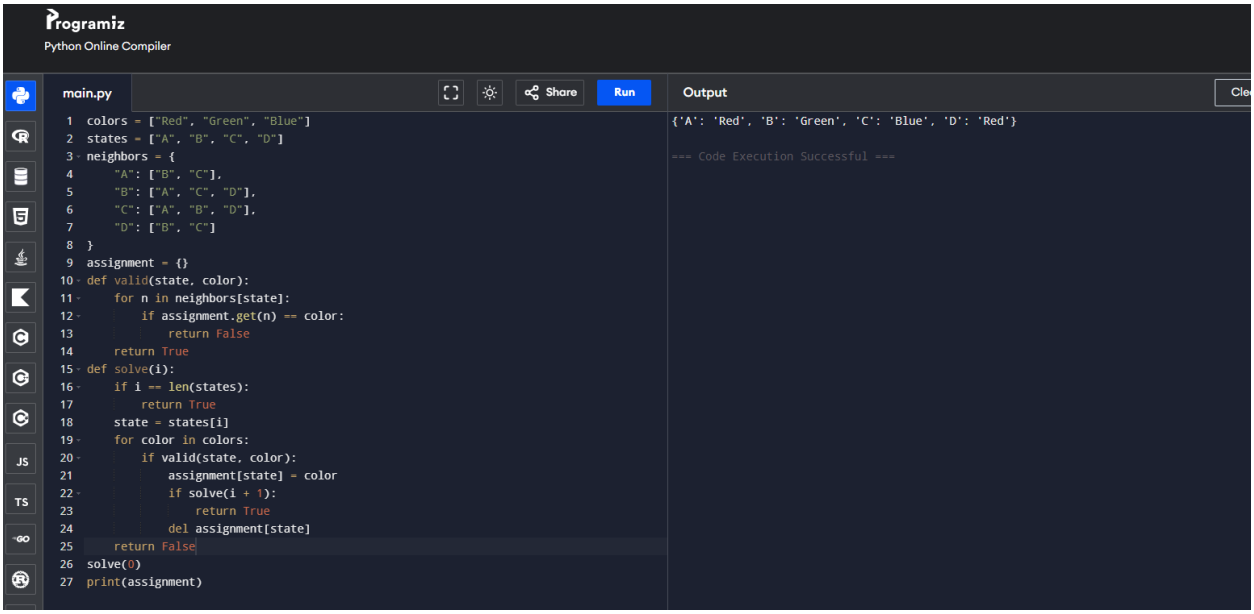
Flowchart


START
Input states, colors, and neighbors
Initialize empty assignment
Select the next unassigned state
Is a state available?
Yes
No
Try a color for the state
Display color assignment
Is the color valid (no same color as neighboring states)?
STOP
Yes
No
Assign the color
Try next color
Solve next state
Solution found?
Yes
No
Return successful
Remove assigned color (Backtrack)
Try another color

Program

```
colors = ["Red", "Green", "Blue"]
states = ["A", "B", "C", "D"]
neighbors = {
   "A": ["B", "C"],
   "B": ["A", "C", "D"],
   "C": ["A", "B", "D"],
   "D": ["B", "C"]
}
assignment = {}
def valid(state, color):
   for n in neighbors[state]:
      if assignment.get(n) == color:
         return False
   return True
def solve(i):
   if i == len(states):
      return True
   state = states[i]
   for color in colors:
      if valid(state, color):
         assignment[state] = color
         if solve(i + 1):
            return True
         del assignment[state]
   return False
solve(0)
print(assignment)
```

Output



Result

The map was successfully colored without violating any constraints.

Conclusion

The CSP approach successfully assigns colors so that no two adjacent regions share the same color.